



DX-LR22-433T22D

Serial port

application guide

Version: 2.0

Date: 2025-5-12



Updated records

Version	Date	Instructions	Author
V1.0	2024/12/12	Initial version	SML
V1.1	2025/03/18	Optimization instructions	SML
V2.0	2025/05/12	Update Instruction parameters	SML

Contact Us

SHEN ZHEN DX-SMART TECHNOLOGY CO.,LTD,

Email: manager@szdx-smart.com

Tel: 0755-2997 8125

Whatsapp: +86 15798463070

Website: en.szdx-smart.com

Address: 601, A1 Block, Huafengzhigu, Hang Kong Road, Hang Cheng Street, Baoan District, Shenzhen

Contents

1. Introduction.....	- 6 -
1.1. Basic parameters of serial port.....	- 6 -
1.2. Module default RF basic parameters.....	- 6 -
1.3. Transmission mode and AT command mode.....	- 6 -
2. PC side test tool.....	- 7 -
2.1. Pc-side testing software.....	- 7 -
3. Serial port usage.....	- 8 -
3.1. Module test minimum system.....	- 8 -
3.2. Notes for module use.....	- 8 -
3.3. Example of module use operation.....	- 9 -
3.3.1. Module to module transparent transfer.....	- 9 -
3.3.2. Module and module fixed-point transmission.....	- 9 -
3.3.3. Module and module broadcast transmission.....	- 10 -
3.3.4. Mode switching.....	- 11 -
3.3.4.1. Mode 1 (Configure working mode via AT command).....	- 11 -
3.3.4.2. Method 2 (Configure four working modes via M0/M1 pin).....	- 12 -
3.3.5. Instructions on the AUX module working state indicator foot.....	- 12 -
4. Related AT commands are explained in detail.....	- 13 -
4.1. Command Format Description.....	- 13 -

4.2. Response Format Description	- 13 -
4.3. An example of the AT command.....	- 14 -
4.4. List of AT commands	- 14 -
5. AT command details	- 15 -
5.1. Basic commands.....	- 15 -
5.1.1. Enter or exit the AT command mode.....	- 15 -
5.1.2. Test the instruction	- 15 -
5.1.3. Software restart.....	- 15 -
5.1.4. factory data reset.....	- 16 -
5.2. Serial port parameters.....	- 16 -
5.2.1. Set \ query-serial port baud rate.....	- 16 -
5.2.2. Set \ Query - serial port check bit.....	- 16 -
5.3. LORA parameters.....	- 17 -
5.3.1. Query configuration information	- 17 -
5.3.2. Settings \ Query - One click to configure module air rate and communication distance	- 18 -
5.3.3. Set \ query-transfer mode.....	- 19 -
5.3.4. Set \ Query - working mode.....	- 21 -
5.3.5. Settings \ Query - Hardware control pin status	- 22 -
5.3.6. Set \ Query - working channel.....	- 22 -
5.3.7. Settings \ Query - Device address	- 24 -



5.3.8. Settings \ Query - Module Key switch	- 25 -
5.3.9. Setup - Module key.....	- 25 -
5.3.10. Set \ query-subpacket length	- 25 -
5.3.11. Settings \ Query - Packet RSSI.....	- 26 -
5.3.12. Set \ Query - Transmit power	- 26 -
5.3.13. Set \ query -LBT status.....	- 27 -
5.3.14. Set \ query-lbt listening threshold	- 27 -
5.3.15. Query - Current channel noise level.....	- 28 -
5.4. List of error codes.....	- 28 -
6. Value-added services.....	- 28 -

Image Index

Figure 1: Computer side serial port software Figure.....	- 7 -
Figure 2: Modular minimal system Figure.....	- 8 -



1. Introduction

DX-LR22-433T22D is a low-power LoRa module, which is built by Shenzhen DX-SMART Technology Co., Ltd. for intelligent wireless data transmission. It uses LLCC68 chip, and integrates the maximum output of 22dBm power amplifier inside the chip, which further improves anti-interference performance and communication distance. It has the advantages of low power consumption, high performance and long distance.

1.1. Basic parameters of serial port

- Default parameters of module serial port: 9600bps/8/n/1 (baud rate/data bit/no check/stop bit)

1.2. Module default RF basic parameters

- Module Rate Class: LEVEL2 (2148bps)
- Module mode switching mode: Mode 1 (AT command control)
- Module subpacket length: 230 bytes
- Module packet RSSI: off
- Module key: 12345
- Module frequency band: 433.15MHz
- Module transmit power: 22dBm
- Module working mode: transparent transmission
- Module power consumption mode: high aging mode
- Module address: ffff
- Module CAD detection time: 4s

1.3. Transmission mode and AT command mode

- Transfer mode: When the module is powered on, it is in transfer mode, at which time it can start transferring data.
- AT command mode: In transmission mode, use "+++" to switch to AT command mode,



which can respond to AT commands. To enter the transmission mode, you need to send "+++" to exit the AT command mode.

2. PC side test tool

2.1. Pc-side testing software

For the computer test software, please download and install sscom5.13.1 computer serial port software in the data package for testing. The interface of the serial port software is as follows:

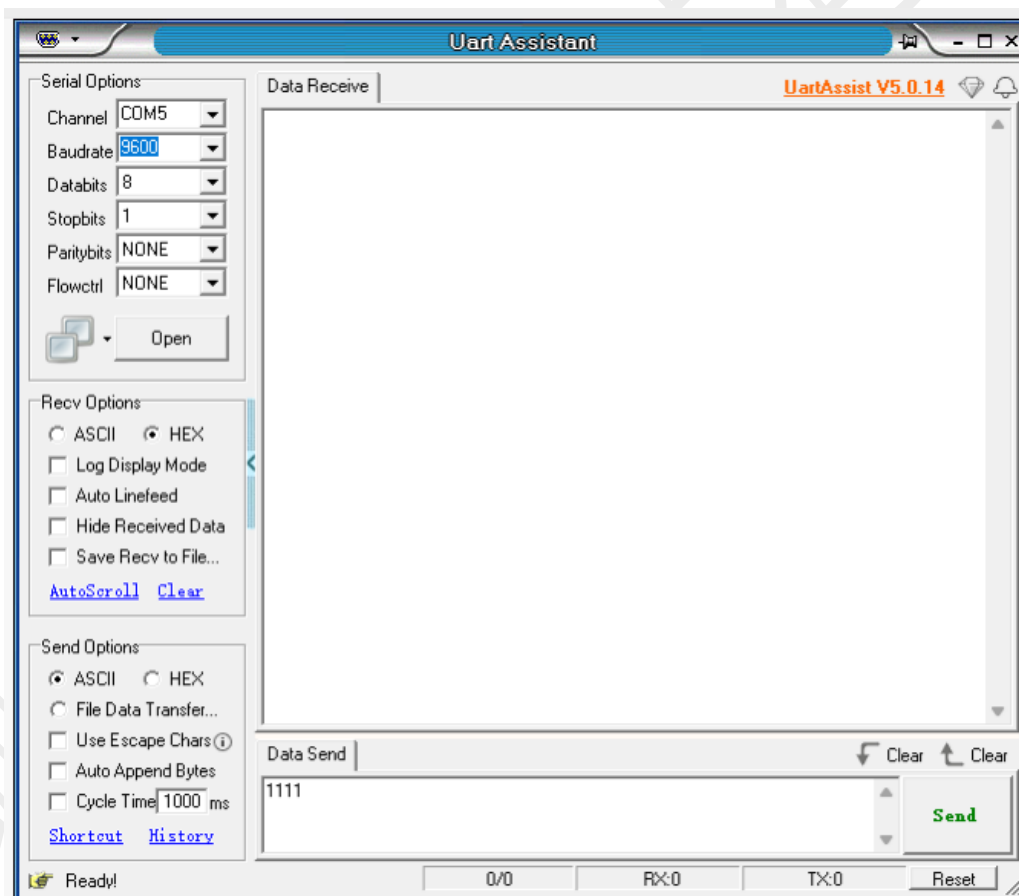


Figure 1: Computer side serial port software diagram1

3. Serial port usage

3.1. Module test minimum system

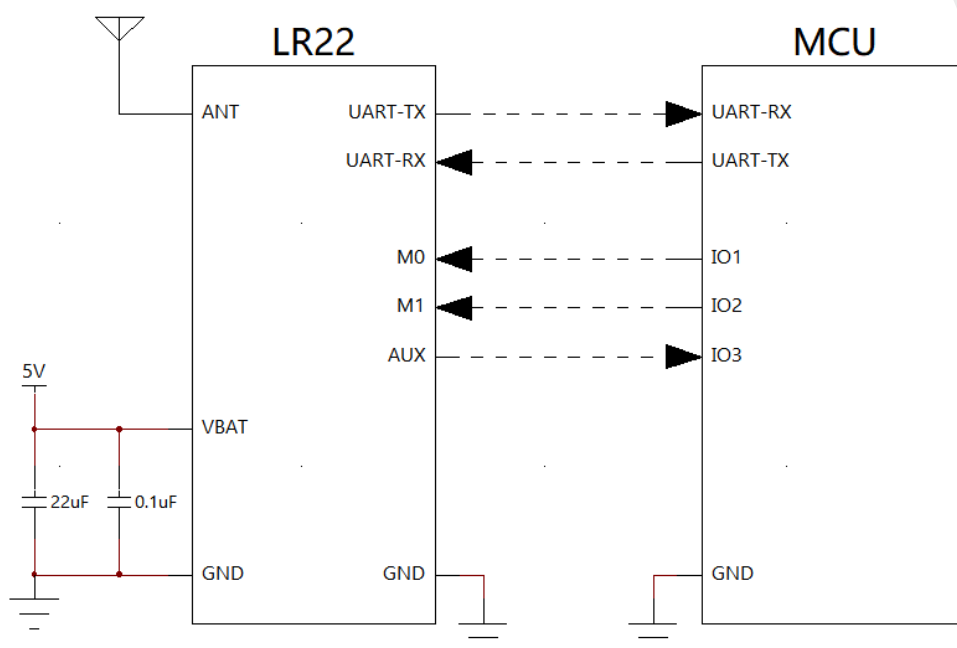


Figure 2: Module minimum system diagram2

3.2. Notes for module use

- Due to the characteristics of LoRa modulation mode, the information transmission delay is long, it is recommended that customers do not transmit large amounts of data at low airspeed, so as to avoid communication anomalies caused by data loss caused by data accumulation.
- When the module is switched from other modes to sleep mode, if there is data that has not been processed, the module will process these data (including receiving and sending) before entering sleep mode.
- When switching from sleep mode to other modes, the module will reconfigure the parameters. During the configuration process, AUX will keep the high level; After



finishing, the output is low, so it is recommended that the user check the AUX level change.

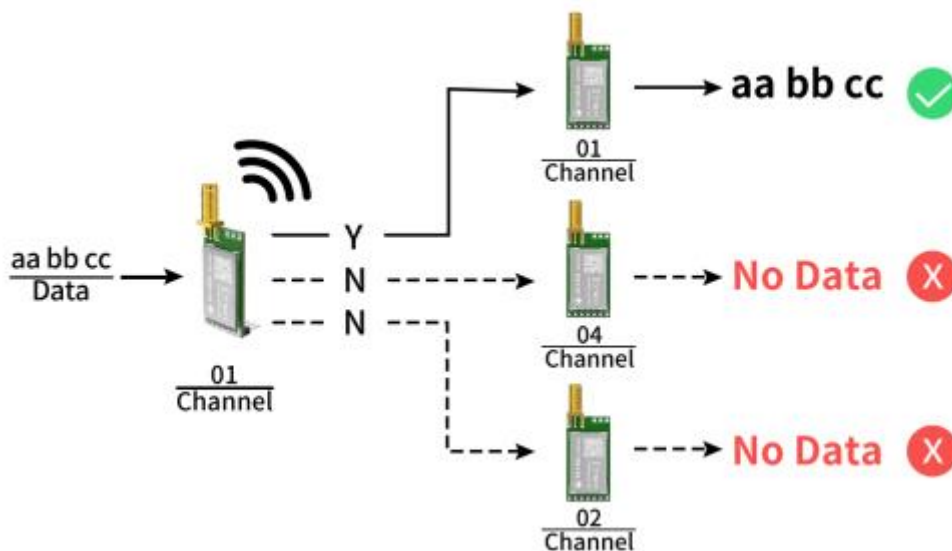
3.3. Example of module use operation

3.3.1. Module to module transparent transfer

Take two DX-LR22-433T22D modules as an example, which are module a and module b

1. Module a and b are connected to the serial port and power supply
2. Module a and b enter AT mode: +++
3. Modules a and b are set to transparent transmission mode: AT+MODE0
4. Modules a and b are configured with the same rate level: AT+LEVEL2
5. Modules a and b are configured with the same channel: AT+CHANNEL01
6. Module a and b restart instruction takes effect: AT+RESET
7. Data transfer: One module sends data, and the other module can receive data

Note: (1) lora is a half-duplex protocol, so only one module can send data at a time
(2) Transparent transmission mode requires the same rate level and channel of two modules to transmit data to each other



3.3.2. Module and module fixed-point transmission

Take two DX-LR22-433T22D modules as an example, which are module a and module b

1. Module a and b are connected to the serial port and power supply



2. Module a and b enter AT mode: +++
3. Modules a and b are set to fixed-point transmission mode: AT+MODE1
4. Modules a and b are configured with the same rate level: AT+LEVEL2
5. Module a and b restart command takes effect: AT+RESET
6. Data transmission: The two modules choose HEX format to send, and transfer data to each other in accordance with the specific data format

Note: (1) lora is a half-duplex protocol, so only one module can be sent at a time

(2) The fixed-point transmission mode is a private protocol, which requires the same rate level of the two modules and the following data format in order to receive and send data normally:

Data format: receiver address (hexadecimal, two bytes) + receiver channel (hexadecimal, one byte) + data (hexadecimal)

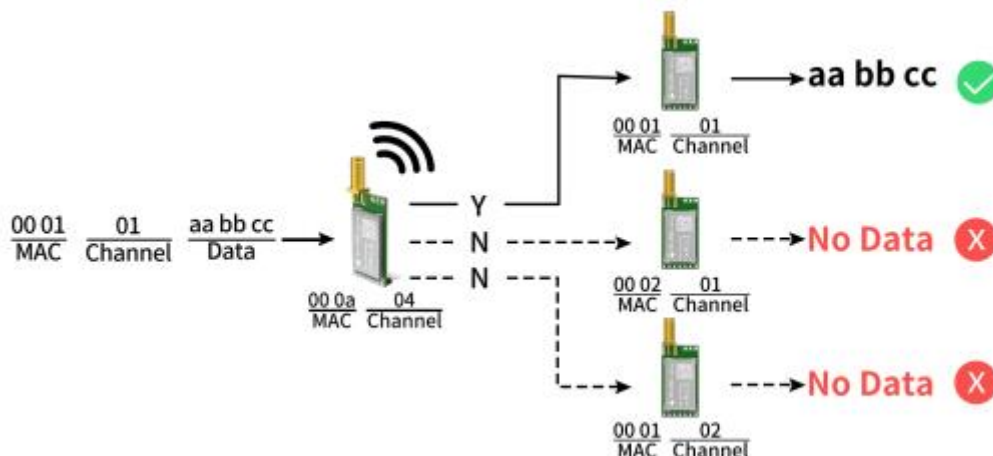
For example:

(Module a as sender; Module b as receiver, module b with address 0001 and channel 01)

Module a sends data to module b aabbcc:

(1) Module a selects HEX format to send

(2) Module a sends the data content as: 00 01 01 61 61 62 62 63 63



3.3.3. Module and module broadcast transmission

Take two DX-LR22-433T22D modules as an example, which are module a and module b

1. Module a and b are connected to the serial port and power supply
2. Module a and b enter AT mode: +++
3. Modules a and b are set to broadcast transmission mode: AT+MODE2
4. Modules a and b are configured with the same rate level: AT+LEVEL2
5. Module a and b restart command takes effect: AT+RESET
6. Data transmission: The two modules choose HEX format to send, and transfer data to each

other in accordance with the specific data format

Note: (1) lora is a half-duplex protocol, so only one module can be sent at a time

(2) The broadcast transmission mode is a private protocol, which requires the same rate level of the two modules and the following data format in order to receive and send data normally:

Data format: receiver channel (hexadecimal, 1 byte) + data (hexadecimal)

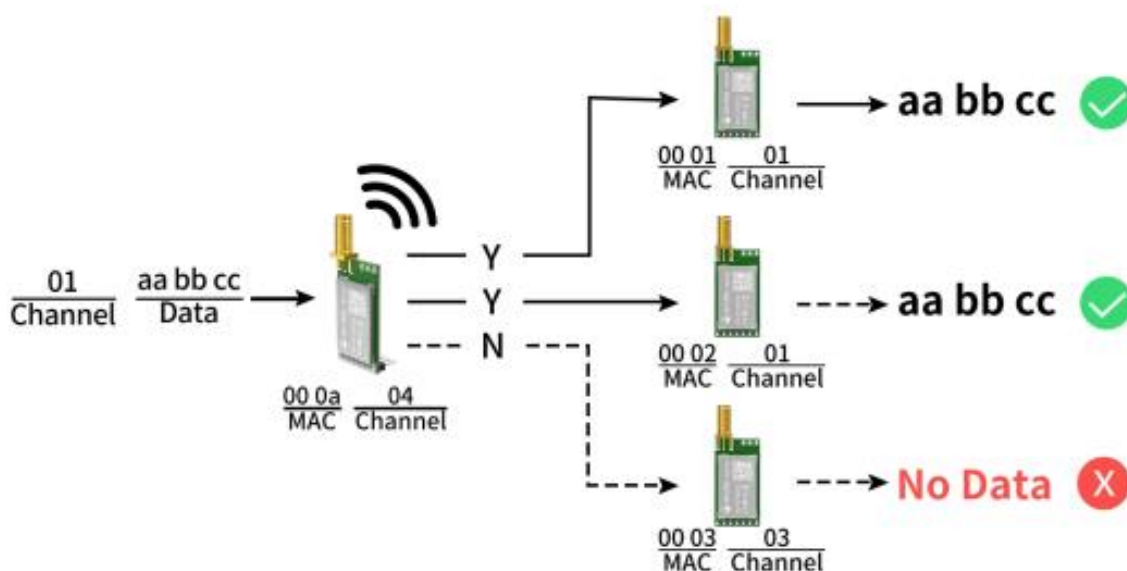
Examples:

(Module a as the sender; Module b as the receiver, module b channel 01)

Module a sends data aabbcc to module b:

(1) Module a selects HEX format to send

(2) Module a sends the data content as: 01 61 61 62 62 63 63



3.3.4. Mode switching

3.3.4.1. Mode 1 (Configure working mode via AT command)

+++ Enter the AT command mode and configure the desired mode with the following command:

- High Aging mode: AT+SLEEP2
- Air Wake Mode: AT+SLEEP1
- Sleep mode: AT+SLEEP0

3.3.4.2. Method 2 (Configure four working modes via M0/M1 pin)

When the module command AT+SWITCH=1, the working mode of the module can be switched through the M0/M1 pin, and the working mode corresponds to the following table:

M0 input level	M1 input level	Operating mode
High level	High level	Sleep mode
High level	Low level	Air wake-up mode
Low level	High level	AT mode
Low level	Low power level	High aging mode

Notes

- 1、Instructions AT+SWITCH are detailed in 5.3.5 Settings \ Query - Hardware Control Pin Status.
- 2、When the instruction AT+SWITCH=1, the internal M0/M1 pin is weak pull-up. When switching the working mode of the module through the M0/M1 pin, please be sure to avoid the pin hanging to prevent the abnormal mode switching.

3.3.5. Instructions on the AUX module working state indicator foot

Pin name	Pin number	Description	Pin output level	Instructions
AUX	5	Indication of module working status	High level	Data transmitting/data receiving/working mode switching
			Low level	Data sending is completed/data receiving is completed/work mode switching is completed

Remarks

When the module detects the received data, AUX will output a high level 2-3ms in advance, which is used to prompt the external MCU to be ready to receive the data.



4. Related AT commands are explained in detail

4.1. Command Format Description

AT+Command<param1, param2, param3> <CR><LF>

- All commands begin with AT and end with <CR><LF>. In the table that shows commands and responses in this document, <CR><LF> is omitted and only commands and responses are displayed.
- All AT command characters are capitalized in English.
- <> is optional content, if there are multiple arguments in the command, separated by a comma ", ", the actual command does not contain Angle brackets.
- <CR> is the carriage return character \r, which is 0X0D in hexadecimal.
- <LF> is the newline character \n, which is 0X0A in hexadecimal.
- If the instruction is executed successfully, the corresponding command is returned with the end of OK, and if it fails, EEROR=<> is returned with the corresponding error code in "<>" (please refer to 5.4).

4.2. Response Format Description

+Indication<=param1, param2, param3> <CR> <LF>

- The response instruction begins with the plus sign "+" and ends with <CR> <LF>
- Equals "=" followed by the response argument
- If there are multiple arguments in the response, they are separated by a comma ", "

4.3. An example of the AT command

Example: Modify the LoRa device baud rate to 9600

Send: AT+BAUD3

Return: OK

4.4. List of AT commands

Instructions	Functions	Instructions
+++	Enter or exit the AT command mode	Power on defaults to transmission mode
AT	Test instructions	Used to test the serial port
AT+RESET	Software restart	-
AT+DEFAULT	factory data reset	-
AT+BAUD	Set \ Query serial port baud rate	Default: 3 (9600)
AT+PARI	Set \ Query serial port check bits	Default: 0 (no parity)
AT+HELP	Query configuration information	-
AT+LEVEL	Set \ Query module air rate and communication distance	Default: 2
AT+MODE	Set \ Query transmission mode	Default: 0 (transparent transfer)
AT+SLEEP	Set \ Query work mode	Default: 2 (high aging mode)
AT+SWITCH	Set \ Query hardware control pin status	Default: 0 (off)
AT+CHANNEL	Set \ Query working channel	Default: 00
AT+MAC	Set \ Query device address	Default: ff,ff
AT+OPENKEY	Set \ Query module key switch	Default: 1 (on)
AT+KEY	Setting the module key	Default: 12345
AT+PACKET	Set \ query subpacket length	Default: 3 (230 bytes)
AT+DRSSI	Set \ Query packet RSSI	Default: 0 (off)
AT+POWE	Set \ Query transmit power	Default: 22
AT+LBT	Set \ Query LBT status	Default: 0 (off)
AT+LRSSI	Set \ query LBT listening threshold	Default: -100

AT+ERSSI Query the current channel noise level -

5. AT command details

5.1. Basic commands

5.1.1. Enter or exit the AT command mode

Features	Instructions	Response	Instructions
Enter or exit the AT command mode	+++	Entry AT or Exit AT Power On	Entry AT: Enter the AT command mode Exit AT: Exits the AT command mode Power on defaults to transfer mode

Notes:

- 1、 It will be reset automatically when you exit the AT command mode.
- 2、 The command will not be saved when the power is off.

5.1.2. Test the instruction

Function	Instructions	Response	Instructions
Testing	AT	OK	

5.1.3. Software restart

Features	Instructions	Response	Instructions
Software restart	AT+RESET	OK Power On	

5.1.4. factory data reset

Features	Instructions	Response	Instructions
factory data reset	AT+DEFAULT	OK Power On	

5.2. Serial port parameters

5.2.1. Set \ query-serial port baud rate

Features	Instructions	Response	Instructions
Query baud rate	AT+BAUD	+BAUD= <baud>	<baud> The baud rate corresponds to the serial number
Set baud rate	AT+BAUD<baud>	+BAUD= <baud> OK	1: 2400 5:38,400 2:4800 6:57,600 3:9,600 7:115,200 4:19200 Default: 3(9600)

Notes:

After setting this instruction, it should be restarted to take effect.

5.2.2. Set \ Query - serial port check bit

Features	Instructions	Response	Instructions
Query the serial port check bit	AT+PARI	+PARI= <param>	< param> sequence number
Set the serial port check bit	AT+PARI<param>	+PARI= <param> OK	0: No validation 1: odd check 2: Even check Default: 0

Notes:

After setting this instruction, it should be restarted to take effect.

5.3. LORA parameters

5.3.1. Query configuration information

Features	Instructions	Response	Instructions
Query module Basic configuration Information	AT+HELP	<pre> ===== LoRa Parameter: +VERSION=<version> MODE:<mode> LEVEL:<level> SLEEP:<sleep> Frequency:<frequency> MAC:<mac> Bandwidth:<bandwidth> Spreading Factor:<spreading factor> Coding rate:<coding rate> CRC:<crc> IQ:<iq> Power:<power> ===== </pre>	LoRa Parameter: LoRa parameter <version>: Version <mode> : Transmission mode <level> : Air rate configuration <sleep> : Working mode <frequency> : frequency of work <mac> : device address <bandwidth> : RF bandwidth <spreading factor> : Spreading factor <coding rate> : The RF coding rate <crc> : CRC check <iq> : Whether the IQ signal is flipped <power> : Transmit power

Examples:

Query module basic information

Send: AT+HELP

Return to: =====

```
LoRa Parameter:
+ VERSION = V1.2.1
MODE:0
LEVEL:2 >> 2149bps
SLEEP:2
Frequency:433150000hz >> 00
MAC:ff,ff
Bandwidth:6
Spreading Factor:11
Coding rate:1
CRC:0(false)
IQ:0(false)
Power:22dBm
=====
```

5.3.2. Settings \ Query - One click to configure module air rate and communication distance

Features	Instructions	Response	Instructions
Query module parameters	AT+LEVEL	+LEVEL = <param>	<param> : 0-7, Air speed and communication range configuration, with eight gears
Setting module parameters	AT+LEVEL<param>	+LEVEL = <param> OK	Default value: 2

Notes:

1. You can choose different gears according to your own data volume and communication distance (data volume and distance can refer to the table below). The larger the air character rate, the faster the amount of data that can be sent, the shorter the communication distance. The CHAR rate is inversely proportional to the distance.
- 2, the command will RF bandwidth, RF coding rate, spread spectrum factor has been set, can be used directly.

3. The LEVEL of transmitting equipment and receiving equipment should be consistent to receive and send data.
4. After setting the instruction, it needs to be restarted.

Note: The following table is the configuration parameters under different gears. The following open visible distance is for reference only. The actual distance is subject to the actual measurement.

LEVEL(gear)	SF(spread spectrum factor)	BW(bandwidth KHz)	CR (coding rate)	Air character rate (bit/s)	Open visible distance (Km)
0	11	125	4/8	336	8.0
1	11	250	4/5	1075	-
2	11	500	4/5	2148	-
3	8	250	4/5	6250	-
4	8	500	4/6	10417	-
5	7	500	4/6	18229	-
6	6	500	4/5	37500	-
7	5	500	4/5	62500	-

5.3.3. Set \ query-transfer mode

Features	Instructions	Response	Instructions
Query transmission mode	AT+MODE	+MODE=<param>	param: 0,1,2 0: Transparent transmission
Set transmission mode	AT+MODE<param>	+MODE=<param> OK	1: Fixed-point transmission 2: Broadcast transmission Default setting: 0

Notes:

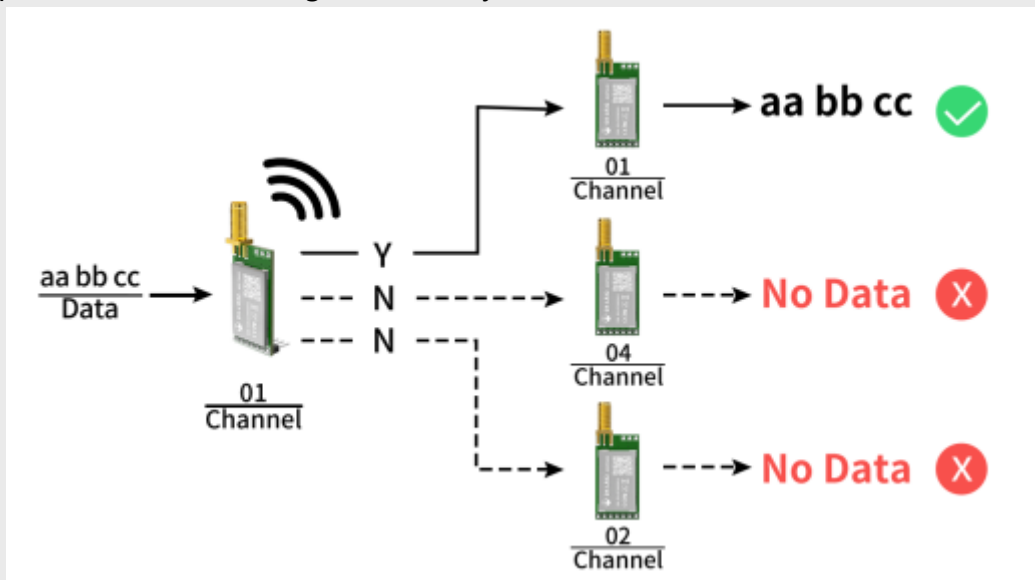
- 1、 After setting this instruction, it should be restarted to take effect.
- 2、 Transparent transmission data format: Send data directly
- 3、 Fixed-point transmission data format: device address (hexadecimal, two bytes) + channel number (hexadecimal, one byte) + data (hexadecimal)



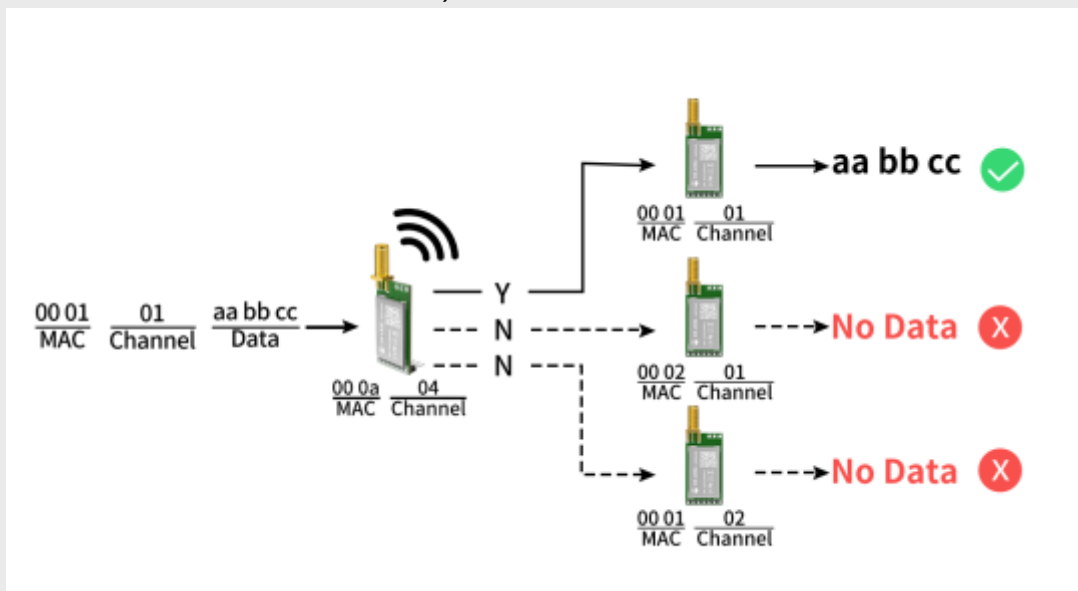
4、Broadcast transmission data format: channel number (hexadecimal, one byte) + data (hexadecimal)

Examples:

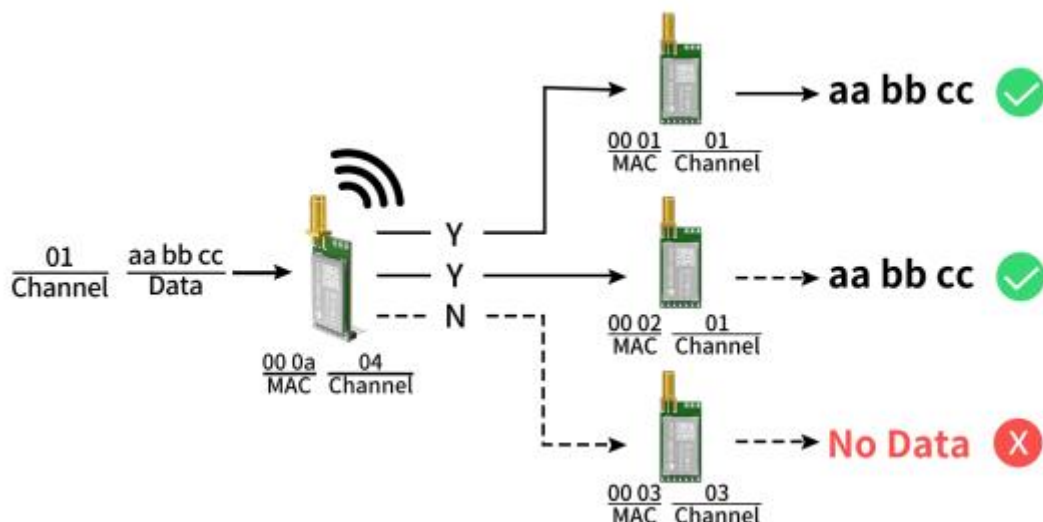
1、Transparent transfer: Sending data directly



2、Fixed-point transmission: the address of the receiving module is 0001, and the channel is 01; The data sent by the transmitting module is aabbcc, then the data content sent is: 000101aabbcc (hexadecimal: 000101 61 61 62 62 63 63)



3、Broadcast transmission: the channel of the receiving module is 01, the data sent by the transmitting module is aabbcc, and the data content is 01aabbcc (hexadecimal: 01 61 61 62 62 63 63).



5.3.4. Set \ Query - working mode

Features	Instructions	Response	Instructions
Query working mode	AT+SLEEP	+SLEEP=<param>	< param> serial number 0: Sleep mode
Set working mode	AT+SLEEP<param>	+SLEEP=<param> OK	1: Air wake-up mode 2: High Time mode Default value: 2

Notes:

1、 Sleep mode: In this mode, both MCU and RF go to sleep.

Method 1: Use the serial port to sleep and wake up, send AT+SLEEP0 to enter the sleep mode, when you wake up, send AT+WAKEUP or any data to wake up, after waking up the module is in the AT command state.

Method two: use M1, M0 foot to sleep and wake up, send AT+SWITCH1 to open M0, M1 mode switch, after the module automatically restarts, set M0=1, M1=1, enter the sleep mode, arbitrarily switch M0, M1, module exit sleep mode.

2. Air wake-up mode:

A. In this mode, the module performs CAD detection in a period of four seconds (the overall sleep time is: 4s minus CAD detection time). If the module detects data, it will enter the receiving mode, and automatically enter the sleep after receiving the data. During the sleep period, the RF will sleep, and the MCU will not sleep.

B. When using the air wakeup mode, both the receiver and the transmitter should be in the air

wakeup mode before they can receive and send data.

C. This mode can be written and saved.

Note: CAD explanation: LoRa CAD (Channel Activity Detection) is a technology used to detect channel activity in LoRa network. It is used to determine whether there is activity (such as transmission by other devices) on a specified physical channel to help the device choose the right time to send and avoid collisions.

3, high aging mode: In this mode, the module is always in the receiving state and can receive data from other devices at any time. When the serial port of the module receives the data from the master control, it switches to the transmitting state, and transmits the data out. After the transmission is completed, it switches back to the receiving state.

4. After setting the command, it needs to be restarted.

5.3.5. Settings \ Query - Hardware control pin status

Features	Instructions	Response	Instructions
Query the hardware control pin status	AT+SWITCH	+SWITCH=<param>	<param> : 0,1 0: close
Sets the hardware control pin state	AT+SWITCH<param>	+SWITCH=<param> OK	1: Open Default: 0 (off)

Notes:

- After setting this instruction, it should be restarted to take effect.
- When SWITCH=1, the working mode can be selected through the module hardware pins M0, M1: (below 0 is low level, 1 is high level)
 - When M0=0, M1=0, the module is in high aging mode;
 - When M0=1, M1=0, the module is in the air wake-up mode;
 - When M0=0, M1=1, the module is in AT mode;
 - When M0=1, M1=1, the module is in sleep mode.
- When the module enters the sleep mode through the pin control, it cannot wake up through the serial port or the command, and must exit the sleep mode through M0 and M1.
- When SWITCH=0, the working mode can be selected by the command AT+SLEEP.

5.3.6. Set \ Query - working channel

Features	Instructions	Response	Instructions
Query the working channel	AT+CHANNEL	+ CHANNEL= <param>	param: 00-63 (hexadecimal)
Set the working channel	AT+CHANNEL <param>	+CHANNEL= <param> OK	Starting at 433.15Mhz, And increase at 1Mhz Default setting: 00

Notes:

1. This module is set with 100 general channels, if you need more, please contact our company.
2. After the instruction is set, it needs to be restarted.
3. When multiple receiving devices are too close to the transmitting device; It may lead to the receiving equipment of different channels can receive data, so the distance between the transmitting equipment and the receiving equipment is as far as possible.

Note: The following table is a comparison of the operating frequency bands of different channels, in Mhz.

Channel	Operating frequency band	Channel	Operating frequency band	Channel	Operating frequency band	Channel	Operating frequency band
00	433.15	19	458.15	32	483.15	4B	508.15
01	434.15	1A	459.15	33	484.15	4C	509.15
02	435.15	1B	460.15	34	485.15	4D	510.15
03	436.15	1C	461.15	35	486.15	4E	511.15
04	437.15	1D	462.15	36	487.15	4F	512.15
05	438.15	1E	463.15	37	488.15	50	513.15
06	439.15	1F	464.15	38	489.15	51	514.15
07	440.15	20	465.15	39	490.15	52	515.15
08	441.15	21	466.15	3A	491.15	53	516.15
09	442.15	22	467.15	3B	492.15	54	517.15
0A	443.15	23	468.15	3C	493.15	55	518.15
0B	444.15	24	469.15	3D	494.15	56	519.15
0C	445.15	25	470.15	3E	495.15	57	520.15
0D	446.15	26	471.15	3F	496.15	58	521.15
0E	447.15	27	472.15	40	497.15	59	522.15



0F	448.15	28	473.15	41	498.15	5A	523.15
10	449.15	29	474.15	42	499.15	5B	524.15
11	450.15	2A	475.15	43	500.15	5C	525.15
12	451.15	2B	476.15	44	501.15	5D	526.15
13	452.15	2C	477.15	45	502.15	5E	527.15
14	453.15	2D	478.15	46	503.15	5F	528.15
15	454.15	2E	479.15	47	504.15	60	529.15
16	455.15	2F	480.15	48	505.15	61	530.15
17	456.15	30	481.15	49	506.15	62	531.15
18	457.15	31	482.15	4A	507.15	63	532.15

5.3.7. Settings \ Query - Device address

Features	Instructions	Response	Instructions
Looking up device addresses	AT+MAC	+MAC= <param>,<param>	param: Hexadecimal, one byte
Set device address	AT+MAC<param>,<param>	+MAC= <param>,<param> OK	Default Settings: ff,ff

Notes:

After setting this instruction, it should be restarted to take effect.

Examples:

Set the module address to 0a01

Send: AT+MAC0a,01

Return: +MAC=0a,01

OK

5.3.8. Settings \ Query - Module Key switch

Features	Instructions	Response	Instructions
Query the module key switch	AT+OPENKEY	+OPENKEY=<param>	<param> 0,1 0: Close key
Set the module key switch	AT+OPENKEY<param>	+OPENKEY=<param> OK	1: Turn key on Default value: 1

Notes:

- 1、 After setting this instruction, it should be restarted to take effect.
- 2、 When AT+OPENKEY=0, the key function can be turned off, but it will not clear the set key.
- 3、 The OPENKEY of both sender and receiver must be consistent, otherwise the sender and receiver will be abnormal.

5.3.9. Setup - Module key

Features	Instructions	Response	Instructions
Set the module key	AT+KEY<param>	+KEY=<param> OK	<param>0~65535 Default value: 12345

Notes:

1. After setting the instruction, it should be restarted to take effect.
2. The instruction can only be set and cannot be queried.
3. The key of the receiver and the sender must be consistent, otherwise the receiver and the sender will be abnormal.
4. If you need to restore the key value to the DEFAULT, you need to send AT+DEFAULT to restore all parameters of the module to the default value.

5.3.10. Set \ query-subpacket length

Features	Instructions	Response	Instructions
Inquire about the subcontract length	AT+PACKET	+ PACKET=<param>	param: 0 to 3 0:32 bytes 1:64 bytes 2:128 bytes 3:230 bytes
Set the subpacket	AT+PACKET<param>	+PACKET=<param>	Default: 3 (230 bytes)

length	OK
--------	----

Notes:

After setting this instruction, it should be restarted to take effect.

5.3.11. Settings \ Query - Packet RSSI

Features	Instructions	Response	Notes
Query packet RSSI	AT+DRSSI	+DRSSI= <param>	<param> : 0,1 0: close 1: Open Default: 0 (off)
Set packet RSSI	AT+DRSSI<param>	+DRSSI= <param> OK	

Notes:

- 1、 After setting this instruction, it should be restarted to take effect.
- 2、 Add the received signal strength to the received packet.

Examples:

Sender: 31 32 33 34 35

Receiver: 31 32 33 34 35 **AB**

Where the last bit "**AB**" is the RSSI value detected in real time, which will change with the strength of the signal.

The RSSI value is calculated as follows (using the value **AB** as an example) :

Actual signal strength = - (0xFF-0xAB) = - (255-171) = -84 dBm

Therefore, the received signal strength for this time is -84 dBm. Note that the RSSI value may be different for each communication, and it depends on the current wireless signal quality. 0xFF in the above formula is a fixed value.

5.3.12. Set \ Query - Transmit power

Features	Instructions	Response	Instructions
Query transmit power	AT+POWE	+POWE= <param>	param: 0-22dBm (take integer values) Default setting: 22dBm
Set transmit power	AT+POWE<param>	+POWE= <param> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

Examples:

Modify the transmit power to 10dBm

Send: AT+POWE10

Return: +POWE=10

OK

5.3.13. Set \ query -LBT status

Features	Instructions	Response	Instructions
Query the LBT status	AT+LBT	+LBT=<param>	<param> : Sequence number 0: closed
Set the LBT status	AT+LBT<param>	+LBT=<param> OK	1: Open Default: 0

Notes:

- 1、 After setting this instruction, it should be restarted to take effect.
- 2、 When the LBT is set to 1, the channel noise will be monitored first. When the noise is greater than the threshold, the channel noise will be monitored continuously, and it will be sent directly after 2S.

5.3.14. Set \ query-lbt listening threshold

Features	Instructions	Response	Instructions
Query the LBT listening threshold	AT+LRSSI	+LRSSI=<param>	<param> : -255 to 0 The current LBT listening threshold
Set the LBT listening threshold	AT+LRSSI<param>	+LRSSI=<param> OK	Default value: -100

Notes:

- 1、 After setting this instruction, it should be restarted to take effect.
- 2、 Changing the parameter will change the LBT listening threshold.

3、 This default value is obtained from our company's test in a specific environment, and the user needs to set the noise level when the channel is idle through AT+ERSSI.

5.3.15. Query - Current channel noise level

Features	Instructions	Response	Instructions
Query the current channel noise level	AT+ERSSI	+ERSSI= <param>	<param> : Current channel noise level

Remarks:

Enquiries only.

5.4. List of error codes

The details of the error code in EEROR=<> are as follows:

Return value	Error Message Description
104	Invalid instructions
105	Invalid parameters
106	Other errors

6. Value-added services

In order to meet the various functional requirements of customers, our company can provide the following technical value-added services:

- Module program customization, such as: IO function port customization, AT instruction customization, etc.
- Module PCB hardware customization, can be customized to the hardware requirements of customer needs.
- A variety of program customization, can be customized according to customer needs, a full set of software and hardware solutions.



-
- A full set of networking solutions customization, can be customized according to customer needs, a full set of networking, gateway solutions.

If you have the above customized requirements, please contact our business personnel directly.